

Problem set 10

Exercise 1

In a self-attention layer, suppose the unnormalized attention score matrix is

$$S = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}.$$

Compute the softmax of S to obtain the normalized attention matrix A .

Solution

We apply the softmax function to each row separately.

Row 1

The first row is

$$(1, 2, 0).$$

Exponentiating each entry gives

$$(e^1, e^2, e^0) = (e, e^2, 1).$$

The sum is

$$e + e^2 + 1.$$

Using the approximations

$$e \approx 2.718, \quad e^2 \approx 7.389,$$

we get

$$e + e^2 + 1 \approx 2.718 + 7.389 + 1 = 11.107.$$

Therefore, the softmax values are

$$\left(\frac{e}{11.107}, \frac{e^2}{11.107}, \frac{1}{11.107} \right) \approx (0.245, 0.665, 0.090).$$

Row 2

The second row is

$$(0, 1, 1).$$

Exponentiating each entry gives

$$(e^0, e^1, e^1) = (1, e, e).$$

The sum is

$$1 + e + e \approx 1 + 2.718 + 2.718 = 6.436.$$

Thus, the softmax values are

$$\left(\frac{1}{6.436}, \frac{e}{6.436}, \frac{e}{6.436} \right) \approx (0.155, 0.422, 0.422).$$

Row 3

The third row is

$$(2, 0, 1).$$

Exponentiating each entry gives

$$(e^2, e^0, e^1) = (e^2, 1, e).$$

The sum is

$$e^2 + 1 + e \approx 7.389 + 1 + 2.718 = 11.107.$$

Hence, the softmax values are

$$\left(\frac{e^2}{11.107}, \frac{1}{11.107}, \frac{e}{11.107} \right) \approx (0.665, 0.090, 0.245).$$

Final Attention Matrix

So, the normalized attention matrix is

$$A = \begin{bmatrix} 0.245 & 0.665 & 0.090 \\ 0.155 & 0.422 & 0.422 \\ 0.665 & 0.090 & 0.245 \end{bmatrix}.$$

Answer

$$A = \begin{bmatrix} 0.245 & 0.665 & 0.090 \\ 0.155 & 0.422 & 0.422 \\ 0.665 & 0.090 & 0.245 \end{bmatrix}$$

Exercise 2

Consider a self-attention layer with the following setup:

- Sequence length: $n = 4$
- Model dimension: $d_{\text{model}} = 8$
- Number of heads: $h = 2$

Assume that for each head:

$$d_k = d_v = \frac{d_{\text{model}}}{h}.$$

Questions:

1. What are the dimensions of Q , K , and V for one head?
2. What is the dimension of the attention score matrix QK^T ?
3. What is the dimension of the softmax attention matrix?
4. What is the dimension of the output of one head (AV)?
5. What is the final output dimension after concatenating all heads?

Solution

Given:

$$n = 4, \quad d_{\text{model}} = 8, \quad h = 2$$

$$d_k = d_v = \frac{8}{2} = 4$$

1. Dimensions of Q , K , V (per head)

$$Q, K, V \in R^{n \times d_k} = R^{4 \times 4}$$

2. Dimension of QK^T

$$QK^T \in R^{4 \times 4} \cdot R^{4 \times 4} = R^{4 \times 4}$$

3. Dimension of attention matrix

After applying softmax row-wise:

$$A \in R^{4 \times 4}$$

4. Dimension of AV

$$AV \in R^{4 \times 4} \cdot R^{4 \times 4} = R^{4 \times 4}$$

5. Final output after concatenation

Each head produces:

$$R^{4 \times 4}$$

Concatenating $h = 2$ heads:

$$\text{Concat} \in R^{4 \times (2 \times 4)} = R^{4 \times 8}$$

Final Answer

- Q, K, V : 4×4
- QK^T : 4×4
- Attention matrix A : 4×4
- Output per head AV : 4×4
- Final output: 4×8